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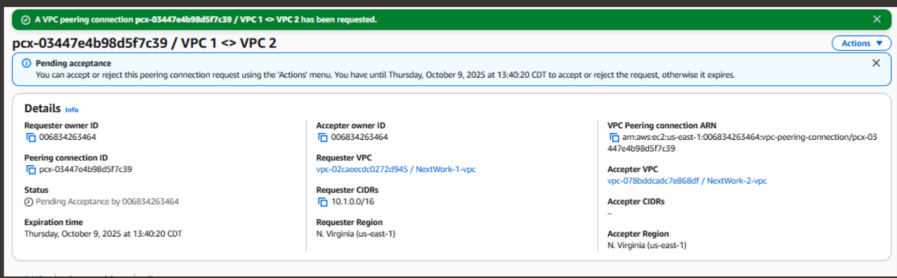
VPC Peering

CH Arpita Chowdhury



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private cloud where you can launch AWS resources in a secure and isolated environment, and it is useful because it gives you control over networking, security, and connectivity.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC for VPC peering.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was creating 2 VPCs and talikng to each other.

This project took me...

This project took me. 2 hours.

In the first part of my project...

Step 1 - Set up my VPC

In this step, I will create two VPCs from scratch and use the visual VPC resource map because I want to set up my VPC.

Step 2 - Create a Peering Connection

In this step, I will set up a connections between VPCs because I want to create a peering connection.

Step 3 - Update Route Tables

In this step, I will set up a way for traffic coming from VPC 1 to get to VPC 2 and Set up a way for traffic coming from VPC 2 to get to VPC 1 because I want to update the route tables.

Step 4 - Launch EC2 Instances

In this step, I will Launch an EC2 instance in each VPC because I want to launch EC2 instances.

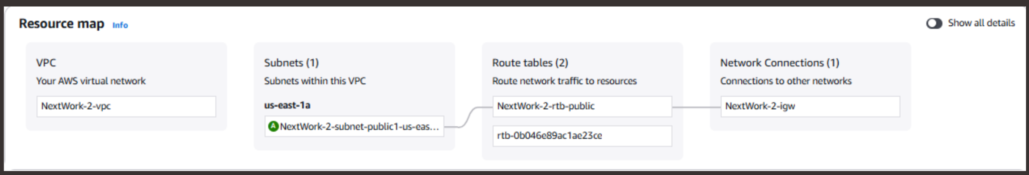
Multi-VPC Architecture

I started my project by launching 2 VPCs and one public subnet for each VPCs.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16 They have to be unique because so the IP addresses of their resources don't overlap. Having overlapping IP addresses could cause routing conflicts and connectivity issues.

I also launched 2 EC2 instances

I didn't set up key pairs for these EC2 instances as AWS actually manages a key pair for us! I don't need to manage key pairs ourselves. Since I've already learnt how to set up key pairs twice in the last two projects, I don't need to do it again this time.

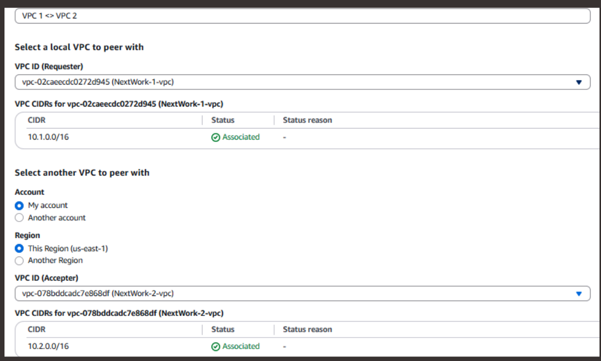


VPC Peering

A VPC peering connection is a direct network connection between two VPCs that enables private communication using AWS’s internal network without requiring internet gateways, VPNs, or dedicated hardware.

VPCs would use peering connections to route traffic privately between them, share resources, and ensure secure, efficient communication without going through the public internet.

The Requester initiates the VPC peering connection, while the Accepter has the authority to approve or reject the connection.



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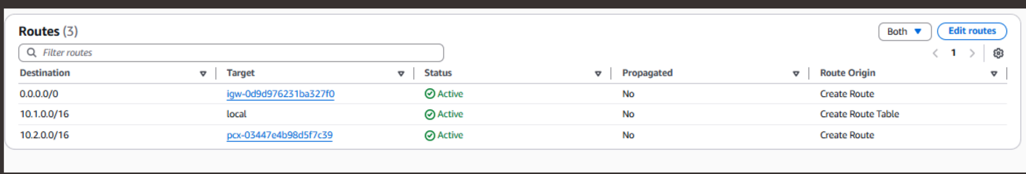
Updating route tables

After accepting a peering connection, my VPCs' route tables need to be updated because traffic in VPC 1 won't know how to get to resources in VPC 2 without a route in your route table! So I need to set up a route that directs traffic bound for VPC 2 to the peering connection I've set up.

My VPCs' new routes have a destination of 10.2.0.0/16. The routes' target was pcx-03447e4b98d5f7c39.



Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0d9d976231ba327f0	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	pcx-03447e4b98d5f7c39	Active	No	Create Route

In the second part of my project...

Step 5 - Use EC2 Instance Connect

In this step, I will use EC2 Instance Connect to connect to connect to the first EC2 instance and fix a connection error because I want connect to Instance 1.

Step 6 - Connect to EC2 Instance 1

In this step, I will use EC2 Instance Connect to connect to Instance 1 and fix another error because I want to connect to Instance 1.

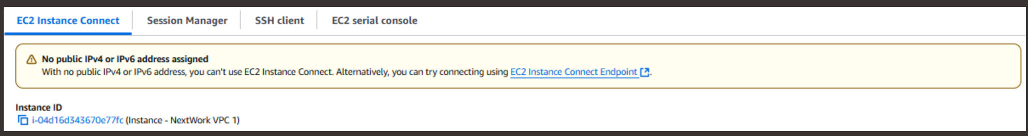
Step 7 - Test VPC Peering

In this step, I will get Instance 1 to send test messages to Instance 2 and solve connection errors until Instance 2 is able to send messages back because I want to test VPC peering.

Troubleshooting Instance Connect

Next, I used EC2 Instance Connect to connect to the first EC2 instance and fix a connection error because I want connect to Instance 1.

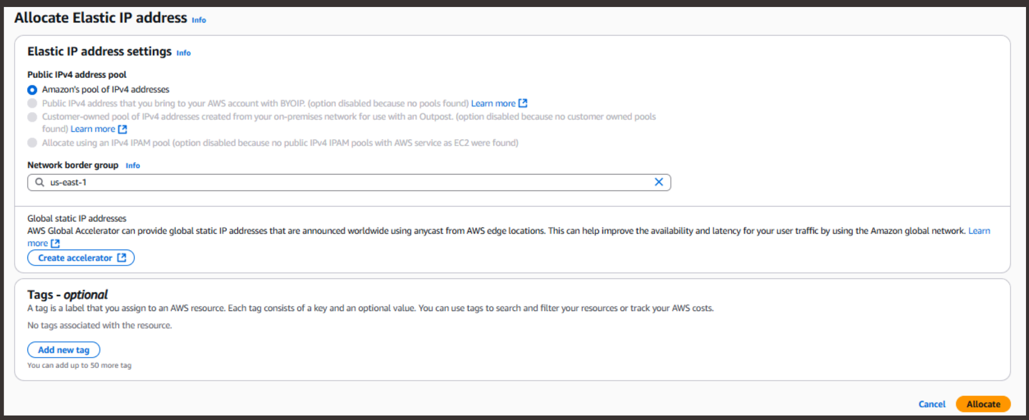
I was stopped from using EC2 Instance Connect as error message: "No public IPv4 address assigned. With no public IPv4 address, you can't use EC2 Instance Connect."



Elastic IP addresses

To resolve this error, I set up Elastic IP addresses. Elastic IP addresses are are static IPv4 addresses that get allocated to your AWS account, and is yours to delegate to an EC2 instance.

Associating an Elastic IP address resolved the error because it gave the EC2 instance a static, unchanging public IPv4 address, ensuring consistent connectivity instead of breaking when the dynamic public IP changed.

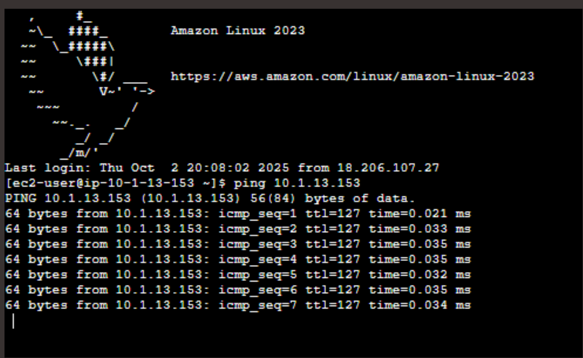


Troubleshooting ping issues

To test VPC peering, I ran the command ping 10.1.13.153.

A successful ping test would validate my VPC peering connection because it validates the test by confirming that different VPCs can communicate with easch other.

I had to update my second EC2 instance's security group because default security group only allows inbound traffic from within the VPC, so traffic from the internet is being cut off. I added a new rule that allows inbound SSH traffic on port 22.



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